

Linear algebra – 5

Bilinear and quadratic forms

12 сентября 2026 года

Theoretical minimum.

Def. A *bilinear form* on V is a function $B(x, y)$ linear in each argument; it is symmetric if $B(x, y) = B(y, x)$. Its matrix in a basis is $(B(e_i, e_j))$, and a change of basis $x = Sx'$ transforms it by *congruence* $B \mapsto S^T B S$. The associated *quadratic form* is $Q(x) = B(x, x)$. The *Gram matrix* of vectors $v_1, \dots, v_k$ of a Euclidean space is $G = (\langle v_i, v_j \rangle)$.

Theorem 1 (law of inertia, Sylvester 1852). Every real quadratic form can be written in a suitable basis as $x_1^2 + \dots + x_p^2 - x_{p+1}^2 - \dots - x_{p+q}^2$, and the pair (p, q) (the *signature*) does not depend on the basis; moreover $p + q = \text{rank } Q$.

Theorem 2 (Sylvester's criterion). A real symmetric matrix A is positive definite if and only if all its leading principal minors are positive; it is positive semidefinite if and only if *all* principal minors are ≥ 0 .

Theorem 3 (simultaneous diagonalisation of a pair of forms). If A is positive definite and B is symmetric, then there is S with $S^T A S = E$ and $S^T B S$ diagonal; the diagonal entries are the roots of $\det(B - \lambda A) = 0$.

Fact 1. $G \succeq 0$ always; $\det G > 0$ exactly when $v_1, \dots, v_k$ are linearly independent, and $\det G$ is the squared volume of the parallelepiped spanned by them.

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1. Reduce $Q(x) = x_1x_2 + x_2x_3 + x_3x_1$ to a sum of squares by Lagrange's method and find its signature. Then prove the law of inertia in general, and show by an example that non-negativity of the *leading* principal minors does not imply positive semidefiniteness.

2. (IMC 2025 shortlist) Let $B: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a bilinear form with matrix A in the standard basis. Assume that for every invertible $P \in GL_n(\mathbb{R})$ the matrices A and $P^T A P$ have the same minimal polynomial. Determine all such bilinear forms B .
3. (Mexmat, 2011) Let A be a real $n \times n$ matrix such that $H = \frac{1}{2}(A + A^T)$ is positive definite. Prove that $\det H \leq \det A$.
4. (ФКН, осень 2020) Square matrices A, B, C of size $n \times n$ satisfy $A^3 = -E$ and $BA^2 + BA = C^6 + C + E$, where C is symmetric. Is $n = 2021$ possible?
5. (IMC) Suppose $a, b, c \in [-1, 1]$ satisfy $1 + 2abc \geq a^2 + b^2 + c^2$. Prove that $1 + 2(abc)^n \geq a^{2n} + b^{2n} + c^{2n}$ for every positive integer n . (*Hint: the hypothesis says that the Gram matrix of three unit vectors with pairwise cosines a, b, c is positive semidefinite.*)